

Gavin Engelstad

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EDUCATION

- Northwestern University**, PhD in Economics 2025 –
- Northwestern University**, MA in Economics 2025 – 2026
- GPA: 3.97/4.0
- Macalester College**, BA in Economics and Mathematics with Honors 2021 – 2025
- GPA: 3.92/4.0
 - *Economics Honors Thesis*: “Cross-Sectional Household Heterogeneity in the Business Cycle”
 - *Mathematics Honors Thesis*: “Topological Data Analysis of Knowledge Networks”

RESEARCH POSITIONS

- Northwestern University**
- Research Assistant for Dr. Matthew Rognlie* June 2026 –
- Maintained and contributed to the Sequence Space Jacobian Python library
- University of California San Diego**
- Research Assistant for Dr. Gaurav Khanna* July 2024 – August 2025
- Cleaned data and implemented econometric models to identify city wage premiums around the globe
- Macalester College and University of Minnesota**
- Research Assistant for Dr. Lori Ziegelmeier and Dr. Russell Funk* May 2024 – August 2025
- Researched how scientific knowledge grows using Topological Data Analysis. Created knowledge networks, optimized persistent homology cycles, and compared results to null models with three other undergrads

WORKING PAPERS

“The Global Value of Cities” (with Aakash Bhalothia, Gaurav Khanna, and Harrison Mitchell)

WORKS IN PROGRESS

- “Topological Data Analysis of Knowledge Networks” (with Tanisha Dodla, Nadezhda Dominguez Salinas, Russell Funk, Frances McConnell, and Lori Ziegelmeier).
- “Bounding Chain Optimal Cycle Representatives in Complex Networks” (with Russell Funk, Frances McConnell, and Lori Ziegelmeier).
- “Distributions of Spaces of Cycle Representatives in Persistent Homology” (with Peter Bubenik, Russell Funk, Dmitriy Morozov, Francis Motta, Jonathan Mousley, Lori Ziegelmeier).

PRESENTATIONS

- North American Meeting of the Urban Economics Association** at the Chicago Fed 2026
- “The Global Value of Cities,” presentation (Upcoming)
- Migration & Organizations Conference** at the Wharton School 2026
- “The Global Value of Cities,” presentation
- STEG Annual Conference** at Strathmore University in Nairobi 2026
- “The Global Value of Cities,” presentation
- Topological Data Visualization Workshop** at the University of Iowa 2025
- “Tutorial on Cycle Optimization,” workshop session
- Algebraic Topology: Methods, Computation, & Science Conference** at Montana State University 2025
- “Bounding Chain Optimal Cycle Representatives in Complex Networks,” poster

UG-Quantitative Methods & Macroeconomics at the Midwest Economics Association Annual Meeting 2025
 “Cross-Sectional Household Heterogeneity in Responses to Macroeconomic Shocks,” presentation

AMS Special Session on Topological Machine Learning at the Joint Mathematics Meetings 2025
 “Topological Insights into the Evolution of Scientific Knowledge,” presentation

PME Undergraduate Student Poster Session at the Joint Mathematics Meetings 2025
 “A Topological Approach to Understanding the Development of Knowledge Networks,” poster (with Frances McConnell)

Academic Summer Showcase Poster Session at Macalester College 2024
 “Topological Data Analysis of Knowledge Networks,” poster (with Lucia Luo)

HONORS AND AWARDS

First-Year Doctoral Studies Fellowship, Northwestern University (2025)
Graduate Research Fellowship Program Honorable Mention, National Science Foundation (2025)
Robert L. Bunting Price in Economics, Macalester College Economics Department (2025)
Best Undergraduate Paper Award, Midwest Economics Association’s 89th Annual Meetings (2025)
DeWitt Wallace Scholarship, Macalester College (2021)
National Merit Scholarship, National Merit Scholarship Cooperation (2021)

TEACHING

Macalester College

- TA for Econ 381: Econometrics (Fall 2023, Spring 2024, Fall 2024)

SKILLS

Software: Python, L^AT_EX, R, Java, Stata, Julia, Mathematica